Blackbird

MWIR InSb detector, 1920 x 1536
10µm pitch, Digital ROIC

Defense Applications



General Description

Blackbird is a state-of-the-art MWIR detector, which is an extension to the High Definition family of detectors at SCD. It includes a 3 Mega-pixel FPA in 1920x1536 pixels format and 10µm pitch. The new FPA is based on SCD's mature InSb technology and on a digital readout circuit implemented in advanced CMOS process. The FPA is very similar in size to SCD's SXGA format Hercules detector and can actually be packaged in the same Dewar. This yields a very large format detector with outstanding image quality, high frame rate, and relatively compact size.

Applications

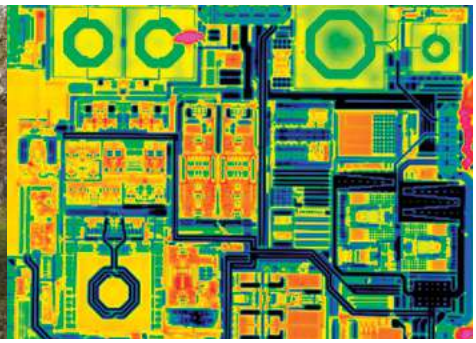
- Persistent surveillance
- Long/medium range Surveillance & Targeting
- IRST
- MWS
- Thermography

Main Features

- High sensitivity: Low readout noise, Low dark current, High Quantum Efficiency
- Range of gain possibilities
- High frame rate: ≥ 90 Hz in full frame
- Simple electronic interface







SemiConductor Devices

YOUR VISION IS OUR VISION



Typical Performance

Parameter	Value
Detector type	InSb 2D array
FPA spectral range	1÷5.4 µm
Format	1920×1536
Pitch	10 µm
Integration modes	ITR, IWR, others
Integration capacitors and their Floor Noise (FN) *	0.4Me- ; FN = 60e- 0.7Me- ; FN = 90e- (ITR mode only) 2.5Me- ; FN = 370e- 4.5Me- ; FN = 950e-
Maximum frame rate (full format, 4 video ports)	≥ 90 F/s
Video output	Digital output, 4 / 2 / 1 ports** (LVDS, 32:4 Channel link)
Digital signal resolution	13 bit
Readout mode	Normal / 2x2 Binning
Readout direction	Bottom-up / Left-Right
Windowing	Flexible: at 2 row steps
NETD (2.5Me- Cap.)	< 25mK at 70% well fill capacity
Local Residual Non Uniformity	0.04% STD/DR at 10-90% well fill capacity
IDCA optical parameters ***	F/3, spectral range: 3.6÷4.9 µm (defined by cold filter)
Cooler options	Full specification: 1W cooler Reduced specification (e.g. 1920×1080 format, 60Hz): 0.75W cooler
Dimensions (with Ricor K543 1W integral rotary cooler)	Weight – approx. 1.1Kg Length (optical axis) – 150 mm

* ITR and IWR modes are possible unless otherwise noted. Floor noise is specified in IWR mode. Floor noise in ITR mode expected to be equal or lower than IWR mode.

** Full frame output is enabled with 1, 2 or 4 video ports. Compared to 4 ports the 2 or 1 video ports enable lower power consumption while max. frame rate is reduced.

*** Other optical configurations are Possible.

Specifications are subject to changes without further notice

SemiConductor Devices

SCD SemiConductor Devices / P.O Box 2250, Haifa, 3102102, Israel
T (972) 4 9902535 / **F** 972 4 9902627 / **www.scd.co.il** / **marketing@scd.co.il**

SCD/UNC/52464